$$\mathcal{C}(\tau) = \sum_{ij} g_i g_j \left\langle \mathrm{e}^{\mathrm{i}\hat{H}_{\mathcal{E}}^{(i)}\tau} \hat{\sigma}_z^{(i)} \mathrm{e}^{-\mathrm{i}\hat{H}_{\mathcal{E}}^{(i)}\tau} \hat{\sigma}_z^{(j)} \right\rangle_{\hat{\rho}_{\mathcal{E}}} \equiv \sum_{ij} g_i g_j \left\langle \hat{\sigma}_z^{(i)}(\tau) \hat{\sigma}_z^{(j)} \right\rangle_{\hat{\rho}_{\mathcal{E}}} . \tag{5.172}$$